Project Synopsis

on

CLOUD-BASED FILE STORAGE AND SHARING SYSTEM

Submitted By-

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1.INTRODUCTION

In today's digital world, a large amount of data such as documents, images, and videos is created every day. Managing and storing this data safely is very important. Traditional storage methods like pen drives and hard disks have limitations such as limited space, risk of data loss, and difficulty in accessing files from different locations.

Cloud storage provides a modern solution by allowing users to store their files on the internet and access them anytime, from anywhere. This project aims to develop a **Cloud-Based File Storage and Sharing System** using FastAPI, AWS S3, and React, where users can easily upload, manage, and share files in a secure and efficient way.

2. Problem Statement

Traditional file storage methods such as pen drives, hard disks, and local systems have limitations like limited storage capacity, risk of data loss, and lack of remote accessibility. Sharing files through these methods is often slow and insecure. Therefore, there is a need for a secure, scalable, and efficient system that allows users to store, access, and share their files anytime and from anywhere, which this project aims to address using a cloud-based solution.

3.Objectives

- To develop a cloud-based system for storing and managing files
- To enable users to upload, download, and delete files easily
- To provide secure user authentication and authorization
- To ensure data safety and reliability using cloud storage (AWS S3)
- To allow users to access files anytime and from anywhere
- To create a simple and user-friendly interface using React

4.Methodology

The project follows a client-server architecture where the frontend is developed using React and the backend is built using FastAPI. The user interacts with the system through a web interface to upload, view, download, or delete files. When a request is made, it is sent to the backend server, which processes the request and handles authentication using JWT.

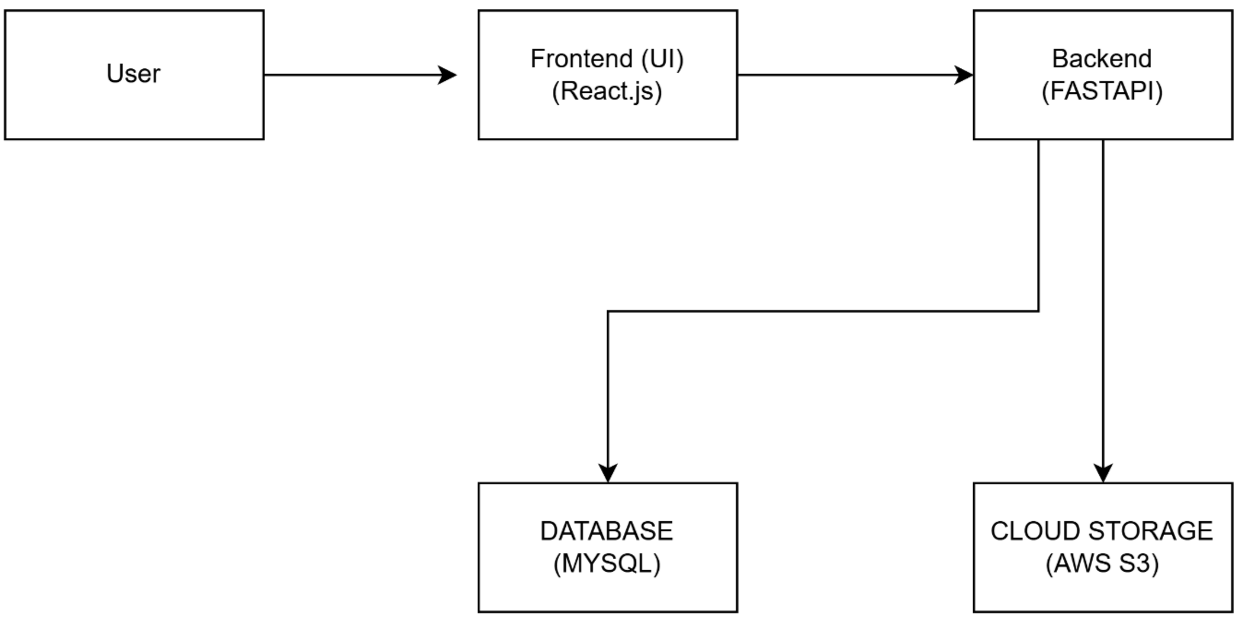
The backend communicates with AWS S3 to store and retrieve files securely, while a database is used to maintain user information and file metadata. The overall workflow includes user authentication, file upload to cloud storage, and file management operations, ensuring efficient and secure handling of data.

5.Tools & Technologies

- **Frontend:** React.js, HTML, Tailwind CSS
- **Backend:** Python, FastAPI
- **Cloud Platform:** AWS S3, AWS EC2
- **Database:** MySQL
- **Authentication:** JSON Web Token (JWT)
- **Version Control:** Git and GitHub
- **Development Tools:** VS Code

6. System Architecture

The system follows a client-server architecture where the user interacts with the frontend built using React. The frontend communicates with the FastAPI backend, which processes requests and stores file data in AWS S3 and metadata in the database. This ensures secure and efficient file management.



7.EXPECTED OUTCOMES

- A secure and functional cloud-based file storage system
- Easy file upload, download, and management features
- Reliable file storage using AWS S3
- User-friendly interface with anytime, anywhere access

8. Applications / Use Cases

- Personal cloud storage for saving and accessing files anytime
- File sharing and collaboration in organizations or teams
- Backup and recovery of important data

9. Future Enhancements

- Add file sharing with links and access permissions
- Implement file versioning and recovery options
- Introduce mobile application support
- Integrate AI-based file search and categorization

10.Conclusion

This project presents a cloud-based file storage and sharing system that provides a secure, scalable, and efficient way to manage digital files. By using technologies like FastAPI, AWS S3, and React, the system allows users to upload, access, and share files easily from anywhere.

Overall, the project improves data accessibility, ensures better security, and offers a user-friendly solution for modern file management needs.